

Jinhong Du

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Target Position: Embodied Intelligence / Robotic Manipulation Algorithm Engineer

Focus: Autonomous manipulation in dense clutter, spanning 3D perception, push-grasp coordination, and real-robot deployment.

Education

South China University of Technology (SCUT) School of Automation Science and Engineering M.Eng. in Electronic Information, Top 10%	Sep. 2024 – Jun. 2027
Jinan University International Energy College B.Eng. in Automation, Top 7%	Sep. 2020 – Jun. 2024

Publications

- **GAPG: Geometry Aware Push-Grasping Synergy for Goal-Oriented Manipulation in Clutter.** **First Author; IEEE ICRA 2026,** Accepted. [\[arXiv\]](#) [\[GitHub\]](#) [\[Demo\]](#)
- **SuperGrasp: Single-View Object Grasping via Superquadric Similarity Matching, Evaluation, and Refinement.** **First Author; IEEE RA-L,** Accepted. [\[arXiv\]](#)

Project Experience

GAPG – Goal-Oriented Push-Grasping in Clutter Lead PointNet++ / PyBullet / Sim2Real	Jun. 2025 – Sep. 2025
• Developed a geometry-aware push-grasp framework for occluded and stacked targets, jointly evaluating grasp feasibility and pushing utility. • Sampled 6D grasp candidates with GraspNet; used PointNet++ to evaluate gripper closing-region geometry and encoded target-surrounding spatial relations for pushing. • Built 40k grasp and 40k pushing samples; achieved zero-shot Sim2Real deployment on ROKAE xMate ER7Pro, with 90.4% grasp success and 95.0% task completion on 15 unseen objects in dense real-world clutter.	
SuperGrasp – Single-View Grasp Candidate Generation, Evaluation, and Refinement Lead PointNet++ / ROS / Sim2Real	Oct. 2025 – Mar. 2026
• Addressed unstable grasping caused by incomplete single-view geometry using superquadric-parameter matching to generate grasp candidates, followed by E-RNet evaluation and refinement. • Built a 1.2k geometric-primitive offline matching library; designed anchor-guided E-RNet to fuse gripper closing-region geometry with surrounding context. • Trained E-RNet on 100k stable grasp samples and deployed zero-shot to ROKAE xMate ER7Pro, achieving 94.12% grasp success, 96.00% task success, and 0.713 s inference time on 10 unseen objects in dense scenes.	
Highly Autonomous Picking for Massive-SKU Warehousing Lead SAC / PyBullet / UR5e Industry Collaboration, Million-RMB Scale	Apr. 2026 – Present
• Extended SuperGrasp to SuperGrasp++ for dense-bin picking by jointly predicting gripper-width corrections and collision risk under severe occlusion, grasp collisions, and wall-adjacent targets. • Developed PushPoint to model spatial relations among the target, neighboring objects, and bin walls; designed the state representation, hybrid discrete-continuous action space, and grasp-score reward, and trained a goal-directed non-prehensile policy with SAC. • Deployed grasping and non-prehensile policies zero-shot to UR5e; on 50+ unseen industrial objects, improved Bin Clear success from 80% with SuperGrasp++ alone to 95% with PushPoint.	

Technical Skills

- **Robot Algorithms:** 3D point-cloud perception, 6D grasping, PointNet / PointNet++, SAC / PPO / TD3, ACT, Diffusion Policy.
- **Simulation & Deployment:** PyBullet, Isaac Gym, ROS, MoveIt, PyTorch, Python; real-robot deployment on UR5e and ROKAE.
- **Other:** English CET-6.